Solving Drop the Handkerchief

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With optimal play, the first Dropper wins about **54.49%** of games under the survival rule below. We compute the values of all **289 million state classes** in a backward pass.

1. The game

In Toshio Sako's *Usogui*, two players take turns as Dropper and Checker. Both players know the past outcomes. Each chooses a second from 1 to 60 without seeing the other's choice.

For each player, s is squandered time (ST), the seconds in their cylinder; t is total time dead (TTD), the seconds from past injections. Write the state as $x = (s_c, t_c, s_d, t_d)$, with Checker first. Both start at $(s, t) = (0, 0)$.

If the Dropper chooses $d \le c$, the Checker succeeds and adds $\ell = c - d + 1$ to ST. At $s_c + \ell \ge 300$, the Checker loses. Otherwise, they swap roles: $x_\ell = (s_d, t_d, s_c + \ell, t_c)$. If $d > c$, the Checker receives an injection of $s_c + 60$ seconds. They survive with probability

$$p(s,t) = \begin{cases} 0.95(1 - s/240)\, 0.75^{t/60}, & s \le 239,\ s + t \le 240, \\ 0, & \text{otherwise.} \end{cases}$$

On revival, ST resets, TTD gains the dose, and the players swap roles: $x_f = (s_d, t_d, 0, t_c + s_c + 60)$. Otherwise, the Checker loses. We choose this probability formula for the model; the manga does not specify it. We exclude the story's extra 61st second.

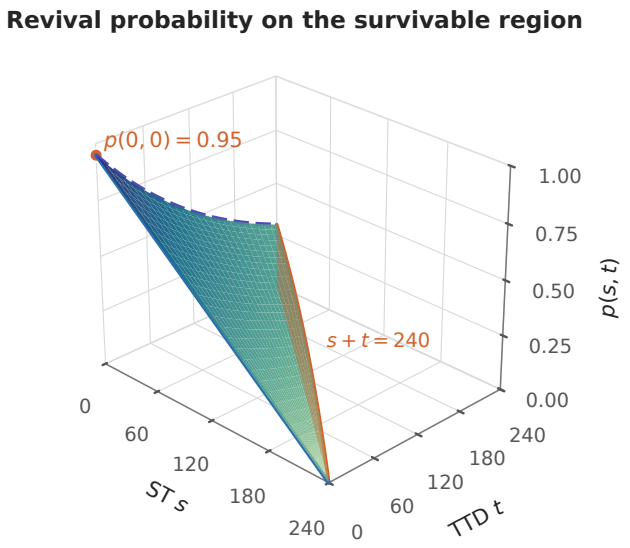


Figure 1: Revival becomes less likely as either time total grows. Beyond $s + t = 240$, the player cannot survive; at $s = 240$, revival also has probability zero.

2. Merge states with the same future

Call a profile *failure-fatal* if its next failed check means death ($p = 0$). We keep its ST and replace its TTD with a common marker $\perp$. This grouping is the *quotient*:

$$\varphi(s,t) = \begin{cases} (s,t), & p(s,t) > 0, \\ (s,\perp), & p(s,t) = 0. \end{cases}$$

Proof. A successful check raises ST, so a failure-fatal player remains failure-fatal. Their exact TTD affects neither the loss on failure nor the grouped state after success. For each action pair, two states in the same group therefore have the same loss probability and next group. Backward induction gives them the same value. QED

From the start, TTD stays in $\{0, 60, 61, \ldots, 300\}$: a revival adds at least 60 and cannot exceed 300. Of $300 \times 242 = 72{,}600$ profiles, we keep $240 + \sum_{t=60}^{240}(241 - t) = 16{,}711$ with $p > 0$, plus 300 markers. For two players, we reduce $72{,}600^2$ states to $17{,}011^2 = 289{,}374{,}121$ classes.

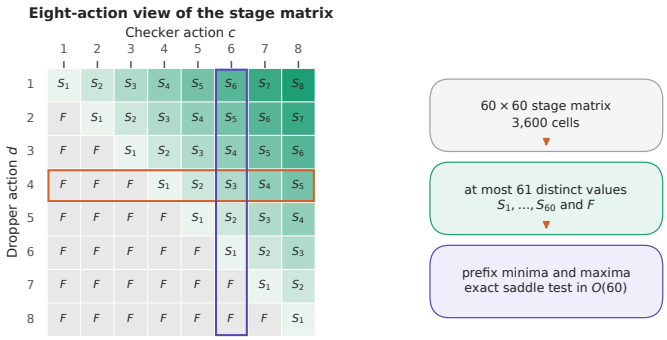


Figure 2: Each diagonal has one success payoff S_ℓ; each failed check has payoff F. We need at most 61 values for the 3,600 action pairs.

3. Solve later states first

We order the classes by a progress score,

$$\Phi = s_c + s_d + \rho_c + \rho_d, \qquad \rho(s,t) = \begin{cases} t, & p > 0, \\ 301, & p = 0. \end{cases}$$

Proof of termination. Success adds $\ell \ge 1$ to Φ, or $\ell + 301 - t > 0$ on becoming failure-fatal. After revival, the increase is 60 if survival remains possible, or $301 - (s + t) \ge 61$ otherwise. Role swaps preserve the score. Since $0 \le \Phi \le 1200$, play ends within 1,201 steps. We solve from layer 1200 down to 0, using known later values. QED

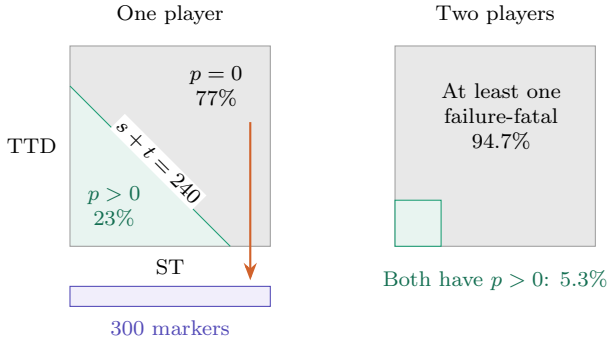


Figure 3: We merge the grey profiles with the same ST. Both players remain alive. The left panel sketches the region; percentages use the allowed TTD values.

4. Calculate both players' strategies

Let V be the current Dropper's win probability minus loss probability. The win probability is $(1+V)/2$. A role swap negates V. With the next states from Section 1, set

$$S_\ell = \begin{cases} 1, & s_c + \ell \geq 300, \\ -V(x_\ell), & \text{otherwise}, \end{cases}$$

$$F = \begin{cases} 1, & p = 0, \\ 1 - p - pV(x_f), & p > 0. \end{cases}$$

For drop second d and check second c, the payoff is

$$M_{d,c} = \begin{cases} S_{c-d+1}, & c \geq d, \\ F, & c < d. \end{cases}$$

The players can randomize their seconds. Their probability vectors a, b give $V = \max_a \min_b a^\mathsf{T} M b$. The finite matrix minimax theorem guarantees this value. We first test fixed seconds: if the best row minimum equals the smallest column maximum, those choices solve the round.

To find a pair that makes the opponent indifferent between seconds, put $D = S_1 - F$ and $\Delta_m = S_{m+1} - S_m$. For $D \neq 0$, compute

$$\boxed{r_1 = 1, \qquad r_k = -\frac{1}{D} \sum_{m=1}^{k-1} \Delta_m r_{k-m} \quad (2 \leq k \leq 60).}$$

If the weights are finite and nonnegative, normalize them: $a_k = r_k / \sum_j r_j$ and $b_k = a_{61-k}$.

Proof of the recurrence. Subtract adjacent rows:

$$(Mb)_d - (Mb)_{d+1} = D b_d + \sum_{m=1}^{60-d} \Delta_m b_{d+m}.$$

Set this to zero and start from $b_{60} \propto 1$ to obtain the recurrence. Column differences give the same weights in reverse order. Thus both players equalize the opponent's payoffs and form an equilibrium. If we clip negative weights to zero, we must check the new pair. QED

5. Check the error

For any probability vectors a, b, set $L = \min_c (a^\mathsf{T} M)_c$ and $U = \max_d (Mb)_d$. The Dropper guarantees L; the Checker limits the payoff to U. Thus $L \leq V \leq U$ for this matrix. We require $U - L \leq 10^{-6}$ and store the midpoint, or a center with a proved error of at most 5×10^{-7}.

Proof of the total bound. A child-value error of at most e changes an expected payoff and the matrix value by at most e. Each step adds at most 5×10^{-7}. Induction over the layers gives $|\widehat{V} - V| \leq (1201 - \Phi) 5 \times 10^{-7}$. QED These are numerical bounds for the stated model; the full-matrix fallback uses floating-point checks.

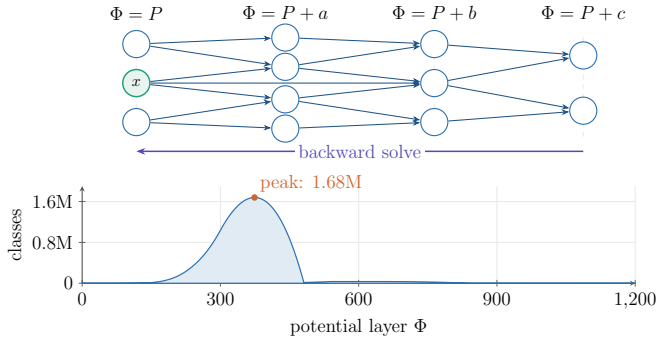


Figure 4: Play moves to higher layers; we solve in reverse. The widest layer contains 1,678,715 classes.

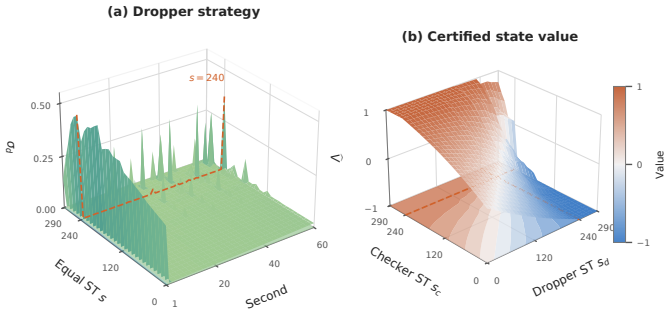


Figure 5: With both TTDs zero: the Dropper's strategy $\sigma_d = a$ at equal ST (left), and the game value as each player's ST changes (right).

6. Results

At the starting state, we obtain

$$V = 0.08985 \pm 0.00061,$$
$$P(\text{first Dropper wins}) = 0.54493 \pm 0.00031.$$

We solve 334,177 classes with fixed actions and 289,039,944 with the recurrence and gap check. No class needs linear programming. We rechecked the saved table at six reference states and 1,200 sampled states; all checks passed.

The repository records a median of **8.01 seconds** for a complete build on an Apple M5 Pro with 15 threads, including verification and saving. We use Python with a C kernel. We bound the full matrix payoffs through a proved shortcut for nonnegative weights and a matrix check for the remaining cases; `src/certificate.md` gives the rounding analysis.

To reproduce the table, run `uv run src/dth.py` from the repository root; run `uv run pytest` for the tests.

Sources. T. Sako, *Usogui*, Shueisha, 2006–2018. J. von Neumann, *Math. Ann.* 100, 295–320 (1928). L. S. Shapley, *PNAS* 39, 1095–1100 (1953).